Jednostrana Z-transf.

→ prije 0 su ∅ $\qquad X[z] = \sum_{n=0}^{\infty} x[n] z^{-n}$

→ ne moramo računati konvergenciju

Primjer:

$$\delta[n] \longrightarrow X[z] = \sum_{n=0}^{\infty} \delta[n] z^{-n} = 1$$

$$\mu[n] \longrightarrow \sum_{n=0}^{\infty} \mu[n] \cdot z^{-n} = \sum_{n=0}^{\infty} (z^{-1})^n = \frac{1}{1-z^{-1}} = \frac{z}{z-1}$$

* kao da je $\alpha^n \mu[n] = 1^n \mu[n]$

$$n\mu[n] \longrightarrow \frac{z}{(z-1)^2}$$

* 6. stranica u formulama

$$n^2 \mu[n] \longrightarrow \frac{z}{(z-1)^3}$$

$$n\alpha^n \mu[n] \longrightarrow \frac{\alpha z}{(z-\alpha)^2}$$

$$\cos(\Omega_0 n)\mu[n] \longrightarrow \frac{1 - z^{-1}\cos(\Omega_0)}{1 - 2z^{-1}\cos\Omega_0 + z^{-2}}$$

SVOJSTVA *na zajedničkim izvodi

1. Linearnost

$$w[n] = \alpha \cdot x[n] \pm \beta \cdot y[n] \longrightarrow W(z) = \sum_{n=0}^{\infty} w[n] \cdot z^{-n} = \sum_{n=0}^{\infty} (\alpha \cdot x[n] \pm \beta \cdot y[n]) z^{-n}$$

$$W(z) = \underbrace{\alpha \cdot \sum_{n=0}^{\infty} x[n] z^{-n}}_{X(z)} \pm \underbrace{\beta \sum_{n=0}^{\infty} y[n] z^{-n}}_{Y(z)} \longrightarrow \boxed{W(z) = \alpha X(z) \pm \beta Y(z)}$$

2. Kašnjenje

$$x[n] \longrightarrow X(z)$$

$$x[n-p] \longrightarrow \sum_{n=0}^{\infty} x[n-p] \cdot z^{-n} = \left| \begin{array}{l} n = m+p \\ n-p = m \\ n=0 \longrightarrow m=-p \\ n=\infty \longrightarrow m=\infty \end{array} \right| = \sum_{m=-p}^{\infty} x[m] \cdot z^{-(m+p)}$$

$$= z^{-p} \sum_{m=-p}^{\infty} x[m] \cdot z^{-m} = z^{-p} \left[\sum_{m=0}^{\infty} x[m] \cdot z^{-m} + \sum_{m=-p}^{-1} x[m] \cdot z^{-m} \right]$$

$$= z^{-p} \cdot X(z) + z^{-p} \sum_{m=-p}^{-1} x[m] \cdot z^{-m}$$

$$x[n-1] \longrightarrow z^{-1} \cdot X(z) + z^{-1} \cdot \sum_{m=-1}^{-1} x[m] \cdot z^{-m} = z^{-1} \cdot X(z) + \cancel{z^{-1}} \cdot x(-1) \cdot \cancel{z^{-(-1)}}$$

$$= z^{-1} \cdot X(z) + x(-1)$$

$$x[n-2] \longrightarrow z^{-2} \cdot X(z) + x(-2) + x(-1) z^{-1}$$

3. Pomak unaprijed

$$x[n] \longrightarrow X(z)$$

$$x[n+p] \longrightarrow z^p \cdot X(z) - z^p \cdot \sum_{m=0}^{p-1} x[m] \cdot z^{-m}$$

$$x[n+1] \longrightarrow z X(z) - z \cdot x(0)$$

$$x[n+2] \longrightarrow z^2 \cdot X(z) - z^2 \cdot x(0) - z \cdot x(1)$$

4. Konvolucija

$$y[n] = h[n] * u[n]$$

$$Y(z) = H(z) \cdot U(z)$$

5. Množenje sa α^n

$$x[n] \longrightarrow X[z]$$

$$\alpha^n x[n] \longrightarrow X\left(\frac{z}{a}\right)$$

$$e^{j\Omega_0 n} x[n] \longrightarrow X\left(z e^{-j\Omega_0}\right)$$

6. Svojstvo modulacije

$$x[n] \cdot \cos(\Omega_0 \cdot n) \longrightarrow \frac{1}{2} X\left(z e^{j\Omega_0}\right) + \frac{1}{2} X\left(z e^{-j\Omega_0}\right)$$

7. Množenje sa n

$$x[n] \longrightarrow X(z)$$

$$n^p x[n] \longrightarrow \left(-z \frac{d}{dz}\right)^p \cdot X(z)$$

$$\hookrightarrow \alpha^n \mu[n] \longrightarrow \frac{z}{z-\alpha}$$

$$n \cdot \alpha^n \mu[n] \longrightarrow -z \frac{d}{dz} \cdot \left(\frac{z}{z-\alpha}\right) = -z \cdot \frac{z'(z-\alpha) - z(z-\alpha)'}{(z-\alpha)^2}$$

$$= -z \frac{\cancel{z}-\alpha - \cancel{z} \cdot 1}{(z-\alpha)^2} = \frac{z\alpha}{(z-\alpha)^2}$$

8. Početna vr. niza

$$x[n] \longrightarrow X(z)$$

$$x[0] = \lim_{z \to \infty} X(z)$$

9. Konačna vrijednost niza

$$\lim_{n \to \infty} x[n] = \lim_{z \to 1} (1 - z^{-1}) \cdot X(z)$$

Kako se iz z domene vratiti u vremensku?

→ rastavimo na parc. razlomke i onda komadić po komadić pomoću tablica

INVERZNA Z-transf.

1°

$$H(z) = \frac{7+5z^{-1}+2z^{-2}}{7+z^{-1}}$$

$$(7+5z^{-1}+2z^{-2}) : (7+z^{-1}) = 1+\frac{4}{7}z^{-1} + \frac{10}{49}z^{-2} \dots$$

$$\begin{array}{l} -7 - z^{-1} \\ \hline \emptyset \quad 4z^{-1}+2z^{-2} \\ \quad -4z^{-1} - \frac{4}{7}z^{-2} \\ \hline \quad \emptyset \quad +\frac{10}{7}z^{-2} \\ \quad\quad -\frac{10}{49}z^{-3} \end{array}$$

$$\rightarrow H(z) = 1+\frac{4}{7}z^{-1}+\frac{10}{49}z^{-2}\dots$$

$$H(z) = \delta[n] + \frac{4}{7}\delta[n-1] + \frac{10}{49}\delta[n-2]\dots$$

$$H(z) = \{\underline{1}, \frac{4}{7}, \frac{10}{49}, \dots\}$$

2°

$$H(z) = \frac{7+5z^{-1}+2z^{-2}}{7+z^{-1}}$$

dijelimo od najvećeg do najmanjeg sad

$$(2z^{-2}+5z^{-1}+7) : (z^{-1}+7) = \underline{2z^{-1}-9}$$

$$\begin{array}{l} -2z^{-2} - 14z^{-1} \\ \hline \emptyset \quad -9z^{-1}+7 \\ \quad +9z^{-1}+63 \\ \hline \quad 70 \end{array}$$

$$\frac{70}{7}\cdot\frac{z}{z+\frac{1}{7}}$$

$$\rightarrow H(z) = -9+2z^{-1}+\frac{70}{7+z^{-1}}$$

$$H(z) = -9+2z^{-1}+10\cdot\frac{z}{z+\frac{1}{7}}$$

$$\rightsquigarrow \boxed{a[n] = -9\delta[n]+2\delta[n-1]+10\left(-\frac{1}{7}\right)^n \mu[n]}$$